Stochastic Financial Models

Part II

Recorded from the lecturer in 2026-2027.

1 Martingales

We work with a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

1.1 Definitions

A *filtration* is a sequence of σ-algebras $(\mathcal{F}_n)_{n\geq 0}$ such that

$$\mathcal{F}_n \subseteq \mathcal{F}_{n+1} \subseteq \mathcal{F}.$$

A *random process* is a sequence $X = (X_n)_{n\geq 0}$ of random variables.
Given a filtration $(\mathcal{F}_n)_{n\geq 0}$ on $(\Omega, \mathcal{F}, \mathbb{P})$, we say that X is a *martingale* if:

(a) X is adapted, i.e. X_n is $\mathcal{F}_n$-measurable for all n.

(b) X is integrable, i.e. X_n is integrable for all n:

$$\mathbb{E}[|X_n|] < \infty.$$

(c) X satisfies the martingale property:

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n \qquad \text{a.s., for all } n.$$

Given (a) and (b), condition (c) is equivalent to

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] = \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n.$$

If (c) is replaced by

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n \qquad \text{a.s., for all } n,$$

or, equivalently,

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] \geq \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n,$$

then $(X_n)_{n\geq 0}$ is a *submartingale*.

If (c) is replaced by

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n \qquad \text{a.s., for all } n,$$

or, equivalently,

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] \leq \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n,$$

then $(X_n)_{n\geq 0}$ is a *supermartingale*.

1.2 Natural filtration

Given a random process $X = (X_n)_{n\geq 0}$, its *natural filtration* $(\mathcal{F}_n^X)_{n\geq 0}$ is defined by

$$\begin{aligned}\mathcal{F}_n^X &= \sigma(X_0, X_1, \ldots, X_n)\\ &= \left\{\{(X_0, X_1, \ldots, X_n) \in B\} : B \in \mathcal{B}(\mathbb{R}^{n+1})\right\}.\end{aligned}$$

This is the smallest filtration making X adapted.

Exercise. Show that a martingale X with respect to $(\mathcal{F}_n)_{n\geq 0}$ is also a martingale with respect to $(\mathcal{F}_n^X)_{n\geq 0}$.

1.3 Examples

Let $X_1, X_2, \ldots$ be a sequence of independent integrable random variables. Set

$$\mathcal{F}_n = \mathcal{F}_n^X.$$

Define

$$S_0 = 0, \qquad Z_0 = 1,$$

and

$$S_n = X_1 + \cdots + X_n, \qquad Z_n = \prod_{i=1}^{n} X_i.$$

- If $\mathbb{E}[X_n] = 0$ for all n, then $(S_n)_{n\geq 0}$ is a martingale.
- If $X_n \geq 0$ and $\mathbb{E}[X_n] = 1$ for all n, then $(Z_n)_{n\geq 0}$ is a martingale.

  To see this, take $A \in \mathcal{F}_n$. Then

$$\begin{aligned}\mathbb{E}[S_{n+1}\mathbf{1}_A] &= \mathbb{E}[X_{n+1}\mathbf{1}_A] + \mathbb{E}[S_n\mathbf{1}_A]\\ &= \underbrace{\mathbb{E}[X_{n+1}]\mathbb{P}(A)}_{=0} + \mathbb{E}[S_n\mathbf{1}_A]\\ &= \mathbb{E}[S_n\mathbf{1}_A].\end{aligned}$$

  Also,

$$\begin{aligned}\mathbb{E}[Z_{n+1}\mathbf{1}_A] &= \mathbb{E}[X_{n+1}Z_n\mathbf{1}_A]\\ &= \underbrace{\mathbb{E}[X_{n+1}]}_{=1}\mathbb{E}[Z_n\mathbf{1}_A]\\ &= \mathbb{E}[Z_n\mathbf{1}_A].\end{aligned}$$

This also shows integrability.

2 Conditional Probabilities and Expectation

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

and

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X\mathbf{1}_B]}{\mathbb{P}(B)}.$$

2.1 Existence and uniqueness

Theorem. Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra. Let X be an integrable random variable. Then there exists a random variable Y such that:

(a) Y is $\mathcal{G}$-measurable;

(b)
$$\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A] \qquad \text{for all } A \in \mathcal{G}.$$

Moreover, if Y' is another random variable satisfying (a) and (b), then

$$Y = Y' \qquad \text{a.s.}$$

Proof of uniqueness. Let X' be another integrable random variable with $X \leq X'$. Suppose Y' satisfies (a) and (b) with respect to X'.

Set
$$A = \{Y > Y'\}, \qquad Z = (Y - Y')\mathbf{1}_A.$$

Then $A \in \mathcal{G}$ and $Z \geq 0$. Hence

$$\begin{aligned}\mathbb{E}[Z] &= \mathbb{E}[(Y - Y')\mathbf{1}_A] \\ &= \mathbb{E}[Y\mathbf{1}_A] - \mathbb{E}[Y'\mathbf{1}_A] \\ &= \mathbb{E}[X\mathbf{1}_A] - \mathbb{E}[X'\mathbf{1}_A] \\ &= \mathbb{E}[(X - X')\mathbf{1}_A] \leq 0.\end{aligned}$$

Therefore $Z = 0$ a.s., so

$$Y \leq Y' \qquad \text{a.s.}$$

Taking $X' = X$, we also obtain $Y \geq Y'$ a.s. Thus

$$Y = Y' \qquad \text{a.s.}$$

2.2 Notation

We write
$$Y = \mathbb{E}[X \mid \mathcal{G}] \qquad \text{a.s.}$$

If $\mathcal{G} = \sigma(W)$, where W is a random variable, we write

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X \mid W].$$

If $X = \mathbf{1}_A$, we write
$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{P}(A \mid \mathcal{G}).$$

End of lecture 1

Optional stopping

We say that a random time

$$T : \Omega \longrightarrow \{0, 1, 2, \ldots\} \cup \{\infty\}$$

is a **stopping time** if

$$\{T \leq n\} \in \mathcal{F}_n \qquad \text{for all } n.$$

Example. Let $(X_n)_{n\geq 0}$ be an adapted process, with values in $\mathbb{Z}$, say. Set

$$T = \inf\{n \geq 0 : X_n = 0\}.$$

Then

$$\{T \leq n\} = \underbrace{\{X_0 = 0\}}_{\in \mathcal{F}_0} \cup \underbrace{\{X_1 = 0\}}_{\in \mathcal{F}_1} \cup \cdots \cup \underbrace{\{X_n = 0\}}_{\in \mathcal{F}_n},$$

with

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_n.$$

So T is a stopping time.

Optional stopping theorem

Let $(M_n)_{n\geq 0}$ be a martingale and let T be a bounded stopping time. Then

$$\mathbb{E}(M_T) = \mathbb{E}(M_0), \qquad \big(M_T(\omega) = M_{T(\omega)}(\omega)\big).$$

Proof. Choose n such that $T \leq n$. Then

$$\begin{aligned} M_T &= M_0 + (M_1 - M_0) + \cdots + (M_T - M_{T-1}) \\ &= M_0 + \sum_{k=1}^{n} (M_k - M_{k-1}) \mathbf{1}_{\{T \geq k\}}. \end{aligned}$$

Now

$$\{T \geq k\} = \{T \leq k-1\}^c \in \mathcal{F}_{k-1},$$

so

$$\mathbb{E}\big((M_k - M_{k-1})\mathbf{1}_{\{T\geq k\}}\big) = 0 \qquad \text{for all } k.$$

Take expectations to see

$$\mathbb{E}(M_T) = \mathbb{E}(M_0) + 0. \qquad \square$$

OST II

Let $(M_n)_{n\geq 0}$ be a martingale and let T be a finite-valued stopping time. Suppose there exists $C < \infty$ such that one of the following holds:

(a)

$$|M_n| \leq C \qquad \text{for all } n \leq T.$$

(b)

$$\mathbb{E}(T) < \infty \qquad \text{and} \qquad |M_n - M_{n-1}| \leq C \qquad \text{for all } n \leq T.$$

Then

$$\mathbb{E}(M_T) = \mathbb{E}(M_0).$$

Proof. Define

$$T_n = T \wedge n.$$

Then T_n is a bounded stopping time. Since $T < \infty$, $T_n = T$ eventually, so

$$M_{T_n} \longrightarrow M_T \qquad \text{as } n \longrightarrow \infty.$$

By optional stopping, we have

$$\mathbb{E}(M_{T_n}) = \mathbb{E}(M_0).$$

If (a) holds, then

$$\sup_n |M_{T_n}| \leq C,$$

so

$$\mathbb{E}(M_{T_n}) \longrightarrow \mathbb{E}(M_T)$$

by bounded convergence.

If (b) holds, then

$$\sup_n |M_{T_n}| \leq CT + |M_0| \qquad \text{and} \qquad \mathbb{E}(T) < \infty,$$

so

$$\mathbb{E}(M_{T_n}) \longrightarrow \mathbb{E}(M_T)$$

by dominated convergence.

In any case, we can let $n \longrightarrow \infty$ to see

$$\mathbb{E}(M_T) = \mathbb{E}(M_0). \qquad \square$$

Examples

Let $X_1, X_2, \ldots$ and $Y_1, Y_2, \ldots$ be independent random variables such that

$$\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = \frac{1}{2}, \qquad \mathbb{P}(Y_n = 0) = \mathbb{P}(Y_n = 2) = \frac{1}{2}.$$

Set

$$S_n = 1 + X_1 + \cdots + X_n, \qquad Z_n = \prod_{i=1}^{n} Y_i, \qquad (Z_0 = 1).$$

Then $(S_n)_{n\geq 0}$ and $(Z_n)_{n\geq 0}$ are martingales.

Set

$$T = \inf\{n \geq 0 : S_n = 0\}.$$

Then

$$\mathbb{P}(T < \infty) = 1,$$

but

$$\mathbb{E}(S_0) = 1, \qquad \mathbb{E}(S_T) = 0.$$

Now set

$$T = \inf\{n \geq 0 : Z_n = 0\}.$$

Then

$$\mathbb{P}(T \leq n) = 1 - 2^{-n},$$

so

$$\mathbb{P}(T < \infty) = 1.$$

In fact,

$$\mathbb{E}(T) < \infty.$$

But

$$\mathbb{E}(Z_T) = 0, \qquad \mathbb{E}(Z_0) = 1.$$

Recall that, for a σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ and an integrable random variable X, $\mathbb{E}(X \mid \mathcal{G})$ is characterized by

$$Y = \mathbb{E}(X \mid \mathcal{G}) \qquad \text{a.s.}$$

if

(a) Y is $\mathcal{G}$-measurable;

(b) Y is integrable and

$$\mathbb{E}(X\mathbf{1}_A) = \mathbb{E}(Y\mathbf{1}_A) \qquad \text{for all } A \in \mathcal{G}.$$

Properties of conditional expectation

(i)

$$\mathbb{E}(\mathbb{E}(X \mid \mathcal{G})) = \mathbb{E}(X).$$

(ii) If X is $\mathcal{G}$-measurable, then

$$\mathbb{E}(X \mid \mathcal{G}) = X \qquad \text{a.s.}$$

(iii) If X is independent of $\mathcal{G}$, then

$$\mathbb{E}(X \mid \mathcal{G}) = \mathbb{E}(X) \qquad \text{a.s.}$$

(iv) If $X \geq 0$ a.s., then

$$\mathbb{E}(X \mid \mathcal{G}) \geq 0 \qquad \text{a.s.}$$

(v) **Linearity.** For W, X integrable and $\alpha, \beta \in \mathbb{R}$,

$$\mathbb{E}(\alpha X + \beta W \mid \mathcal{G}) = \alpha\mathbb{E}(X \mid \mathcal{G}) + \beta\mathbb{E}(W \mid \mathcal{G}).$$

Take

$$Y = \mathbb{E}(X \mid \mathcal{G}), \qquad U = \mathbb{E}(W \mid \mathcal{G}).$$

Then $\alpha Y + \beta U$ is $\mathcal{G}$-measurable and integrable.

For $A \in \mathcal{G}$,

$$\begin{aligned}\mathbb{E}((\alpha X + \beta W)\mathbf{1}_A) &= \alpha\mathbb{E}(X\mathbf{1}_A) + \beta\mathbb{E}(W\mathbf{1}_A) \\ &= \alpha\mathbb{E}(Y\mathbf{1}_A) + \beta\mathbb{E}(U\mathbf{1}_A) \\ &= \mathbb{E}((\alpha Y + \beta U)\mathbf{1}_A)\,.\end{aligned}$$

Hence

$$\alpha Y + \beta U = \mathbb{E}(\alpha X + \beta W \mid \mathcal{G}) \qquad \text{a.s.}$$

Proposition (Tower property)

Let $\mathcal{H}$ be another σ-algebra, with

$$\mathcal{H} \subseteq \mathcal{G}.$$

Then

$$\mathbb{E}(\mathbb{E}(X \mid \mathcal{G}) \mid \mathcal{H}) = \mathbb{E}(X \mid \mathcal{H}).$$

Proof. Take

$$Y = \mathbb{E}(\mathbb{E}(X \mid \mathcal{G}) \mid \mathcal{H}) \qquad \text{a.s.}$$

Then, for $A \in \mathcal{H}$, we have $A \in \mathcal{G}$, so

$$\mathbb{E}(Y\mathbf{1}_A) \underset{A\in\mathcal{H}}{=} \mathbb{E}(\mathbb{E}(X \mid \mathcal{G})\mathbf{1}_A) \underset{A\in\mathcal{G}}{=} \mathbb{E}(X\mathbf{1}_A).$$

Y is integrable and $\mathcal{H}$-measurable, so

$$Y = \mathbb{E}(X \mid \mathcal{H}) \qquad \text{a.s.} \qquad \square$$

End of lecture 2

Properties of conditional expectation (continued)

Proposition (Taking out what is known). Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra, and let X be an integrable random variable. Suppose that Z is a bounded $\mathcal{G}$-measurable random variable. Then

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}] \qquad \text{a.s.}$$

Proof. Take $Y = \mathbb{E}[X \mid \mathcal{G}]$ a.s. Note that ZY is integrable and $\mathcal{G}$-measurable.

Take $A \in \mathcal{G}$. We will show that

$$\mathbb{E}[ZX\mathbf{1}_A] = \mathbb{E}[ZY\mathbf{1}_A],$$

so that ZY satisfies (a) and (b) for ZX, and we are done.

Consider first the case $Z = \mathbf{1}_B$, where $B \in \mathcal{G}$. Then $A \cap B \in \mathcal{G}$, so

$$\begin{aligned}\mathbb{E}[ZX\mathbf{1}_A] &= \mathbb{E}[X\mathbf{1}_{A\cap B}] \\ &= \mathbb{E}[Y\mathbf{1}_{A\cap B}] \\ &= \mathbb{E}[ZY\mathbf{1}_A].\end{aligned}$$

This extends to simple random variables

$$Z = \sum_{i=1}^{n} a_i \mathbf{1}_{B_i}$$

by linearity.

For $Z \geq 0$ and $X \geq 0$, consider

$$Z_n = \left(2^{-n} \lfloor 2^n Z \rfloor\right) \wedge n.$$

Then Z_n is simple, so

$$\mathbb{E}[Z_n X\mathbf{1}_A] = \mathbb{E}[Z_n Y\mathbf{1}_A].$$

Since $X \geq 0$, we have $Y \geq 0$ a.s., so

$$Z_n X \uparrow ZX, \qquad Z_n Y \uparrow ZY.$$

Letting $n \to \infty$,

$$\mathbb{E}[Z_n X \mathbf{1}_A] \longrightarrow \mathbb{E}[ZX\mathbf{1}_A],$$
$$\mathbb{E}[Z_n Y \mathbf{1}_A] \longrightarrow \mathbb{E}[ZY\mathbf{1}_A].$$

Finally, for general X and Z, use

$$X = X^+ - X^-, \qquad Z = Z^+ - Z^-,$$

and linearity to obtain the desired formula. □

Proposition. Let X_1, X_2 be random variables, with X_1 being $\mathcal{G}$-measurable and X_2 independent of $\mathcal{G}$. Suppose that F is a bounded measurable function on $\mathbb{R}^2$. Then

$$\mathbb{E}[F(X_1, X_2) \mid \mathcal{G}] = f(X_1) \qquad \text{a.s.},$$

where

$$f(x) = \mathbb{E}[F(x, X_2)].$$

Recall the optional stopping theorem. Let M be a martingale and let T be a bounded stopping time. Then

$$\mathbb{E}[M_T] = \mathbb{E}[M_0].$$

Theorem. Let M be a martingale and let T be a stopping time. Consider the stopped process

$$M_n^T := M_{T \wedge n}.$$

Then M^T is also a martingale.

Proof. We have

$$M_n^T = M_0 + \sum_{k=1}^{n} (M_k - M_{k-1}) \mathbf{1}_{\{k \le T\}}.$$

Since $\{k \le T\} \in \mathcal{F}_k$, this shows that M_n^T is $\mathcal{F}_n$-measurable. Also, M^T is integrable.

To show the martingale property, it suffices to show, for $A \in \mathcal{F}_n$ and all n, that

$$\mathbb{E}\left[(M_{n+1}^T - M_n^T)\mathbf{1}_A\right] = 0.$$

But

$$M_{n+1}^T - M_n^T = (M_{n+1} - M_n)\mathbf{1}_{\{n+1 \le T\}}.$$

Note that

$$\{n+1 \le T\} = \{T \le n\}^c \in \mathcal{F}_n.$$

So

$$\mathbb{E}\left[(M_{n+1} - M_n) \underbrace{\mathbf{1}_{\{n+1 \le T\}} \mathbf{1}_A}_{\mathcal{F}_n\text{-measurable}}\right] = 0,$$

as required. Note that in the final step we used tower rule and "take out what is known". □

Martingale transforms

Say that a process $(H_n)_{n\geq 1}$ is *previsible* if H_n is $\mathcal{F}_{n-1}$-measurable for all n.

Theorem. Let M be a martingale and let H be a bounded previsible process. We define the transform $H \cdot M$ of M by H by

$$(H \cdot M)_n = Y_n = \sum_{k=1}^{n} H_k(M_k - M_{k-1}), \qquad n \geq 1,$$

and

$$(H \cdot M)_0 = Y_0 = 0.$$

Interpretation. Think of M_n as the price of an asset at time n.
An investor holds H_n units of the asset between time $n-1$ and time n.
Then the profit made from time 0 to time n is Y_n.
The previsible restriction is natural.

Then $H \cdot M$ is also a martingale. In particular,

$$\mathbb{E}[(H \cdot M)_n] = \mathbb{E}[(H \cdot M)_0] = 0.$$

Proof. Note that

$$(H \cdot M)_{n+1} - (H \cdot M)_n = H_{n+1}(M_{n+1} - M_n),$$

and, by taking out what is known,

$$\begin{aligned}\mathbb{E}\left[H_{n+1}(M_{n+1} - M_n) \mid \mathcal{F}_n\right] \\ = H_{n+1} \underbrace{\mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n]}_{=0, \text{ as } M \text{ is a martingale}}.\end{aligned}$$

So $H \cdot M$ is a martingale, as claimed. □

End of lecture 3

2. Pricing contingent claims

2.1 Multi-period asset price model

Let $(\bar{S}_n)_{0\leq n\leq T}$ be a random process in $\mathbb{R}^{d+1}$, with

$$\bar{S}_n = (S_n^0, S_n^1, \ldots, S_n^d) = (S_n^0, S_n), \qquad S_n \in \mathbb{R}^d.$$

We say that $(S_n^0)_{0\leq n\leq T}$ is a *numeraire* if $S_n^0 > 0$ for all n.

If $(S_n^0)_{0\leq n\leq T}$ is a numeraire, we call $(\bar{S}_n)_{0\leq n\leq T}$ a *T-period asset price model*. If $T = 1$, it is called a single-period model.

We think of $(S_n^0)_{0\leq n\leq T}$ as a bank account or bond, and write

$$S_n^0 = (1 + r_n)S_{n-1}^0,$$

where r_n is the interest rate. There are T time periods:

$$0 \longrightarrow 1, \quad 1 \longrightarrow 2, \quad \ldots, \quad T-1 \longrightarrow T.$$

We think of S_n^i as the price of one unit of asset i at time n, for $i = 1, \ldots, d$.

We will focus on *discounted prices*:

$$\bar{X}_n = \frac{\bar{S}_n}{S_n^0} = (1, X_n^1, \ldots, X_n^d), \qquad X_n^i = \frac{S_n^i}{S_n^0}.$$

To determine the model fully, we could suppose that the interest rates r_n are non-random and that prices take discrete values. We would then want to know the values of

$$\mathbb{P}(S_0 = s_0, S_1 = s_1, \ldots, S_T = s_T)$$

for all possible sequences $(s_n)_{0 \le n \le T}$ in $\mathbb{R}^d$.

Portfolios

Let $(\bar{\Theta}_n)_{1 \le n \le T}$ be a previsible random process in $\mathbb{R}^{d+1}$, so that $\bar{\Theta}_n$ is $\mathcal{F}_{n-1}$-measurable. We use $(\bar{\Theta}_n)_{1 \le n \le T}$ to describe a *portfolio* in which we hold Θ_n^i units of asset i during the nth period, from $n-1$ to n.

We say that $(\bar{\Theta}_n)_{1 \le n \le T}$ is *self-financing* if

$$\underbrace{\bar{\Theta}_n \cdot \bar{S}_n}_{\text{value at the end of period } n} = \underbrace{\bar{\Theta}_{n+1} \cdot \bar{S}_n}_{\text{value at the start of period } n+1}, \qquad n = 1, \ldots, T-1.$$

The *value function* of the portfolio, $(V_n)_{0 \le n \le T}$, is given by

$$V_0 = \bar{\Theta}_1 \cdot \bar{X}_0, \qquad V_n = \bar{\Theta}_n \cdot \bar{X}_n = \sum_{i=0}^{d} \Theta_n^i X_n^i, \qquad n = 1, \ldots, T.$$

Proposition. Let $V_0 \in \mathbb{R}$ and let $(\Theta_n)_{1 \le n \le T}$ be a previsible process in $\mathbb{R}^d$. Then there exists a previsible process $(\Theta_n^0)_{1 \le n \le T}$ such that $(\bar{\Theta}_n)_{1 \le n \le T}$ is a self-financing portfolio with initial value V_0, where $\bar{\Theta}_n = (\Theta_n^0, \Theta_n)$. Moreover,

$$V_m = V_0 + \sum_{n=1}^{m} \Theta_n \cdot (X_n - X_{n-1}), \qquad 0 \le m \le T.$$

Proof. We have to satisfy the conditions

$$V_0 = \bar{\Theta}_1 \cdot \bar{X}_0 = \Theta_1^0 + \Theta_1 \cdot X_0$$

and

$$\begin{aligned} \bar{\Theta}_n \cdot \bar{S}_n &= \bar{\Theta}_{n+1} \cdot \bar{S}_n, \\ \Theta_n^0 S_n^0 + \Theta_n \cdot S_n &= \Theta_{n+1}^0 S_n^0 + \Theta_{n+1} \cdot S_n, \qquad n = 1, \ldots, T-1. \end{aligned}$$

We can solve these uniquely for a previsible process $(\Theta_n^0)_{1 \le n \le T}$.

To obtain the formula for V_m, note that the change in discounted value in period n is

$$\sum_{i=1}^{d} \Theta_n^i (X_n^i - X_{n-1}^i) = \Theta_n \cdot (X_n - X_{n-1}).$$

The formula follows by induction. □

2.2 Contingent claims

A *(European) contingent claim* C with maturity T is a non-negative $\mathcal{F}_T$-measurable random variable. We think of C as a contract which pays the investor C at time T. The term "European" signifies that T is fixed.

Here we are thinking of

$$\mathcal{F}_T = \sigma(S_0, S_1, \ldots, S_T).$$

Examples

Restrict to $d = 1$, so there is one risky asset and $S_n \in \mathbb{R}$.

(a) $(S_T - K)^+$ is the payoff of a *call* with strike price K and maturity T. It gives the holder the right, but not the obligation, to buy one unit of the asset at time T for price K.

(b) $(S_T - K)^-$ is the payoff of a *put* with strike price K and maturity T. This is the same as a call, but for selling at price K.

End of lecture 4

Examples of contingent claims (cont.)

$$(\bar{S}_n)_{0 \le n \le T}, \qquad d = 1.$$

Barrier options (B):

(c)
$$C = \begin{cases} (S_T - K)^+ & \text{if } \max\limits_{0 \le n \le T} S_n < B, \\ 0 & \text{otherwise.} \end{cases}$$

Up-and-out call knocked out at B.

(d)
$$C = \begin{cases} (S_T - K)^- & \text{if } \min\limits_{0 \le n \le T} S_n \le B, \\ 0 & \text{otherwise.} \end{cases}$$

Down-and-in put knocked in at B.

2.3 Equivalent probability measures

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Say that a prob. meas. $\widetilde{\mathbb{P}}$ on $(\Omega, \mathcal{F})$ has *density* ρ w.r. to $\mathbb{P}$ if ρ is a random variable, $\rho \ge 0$ and

$$\widetilde{\mathbb{P}}(A) = \mathbb{E}(\rho \mathbf{1}_A), \qquad A \in \mathcal{F}.$$

By standard arguments in this case

$$\widetilde{\mathbb{E}}(X) = \mathbb{E}(\rho X), \qquad \text{any } X \ge 0 \text{ r.v.}$$

In the case where ρ takes values in $(0, \infty)$ then $\mathbb{P}$ has density $1/\rho$ w.r. to $\widetilde{\mathbb{P}}$,

$$\mathbb{P}(A) = \mathbb{E}\left(\rho \frac{\mathbf{1}_A}{\rho}\right) = \widetilde{\mathbb{E}}\left(\frac{1}{\rho}\mathbf{1}_A\right).$$

We write

$$\rho = \frac{d\widetilde{\mathbb{P}}}{d\mathbb{P}} \quad \text{a.s.}, \qquad \frac{1}{\rho} = \frac{d\mathbb{P}}{d\widetilde{\mathbb{P}}} \quad \text{a.s.}$$

Suppose ρ, ρ' are both densities for $\widetilde{\mathbb{P}}$ w.r. to $\mathbb{P}$, then $\rho = \rho'$ a.s.

Say that $\mathbb{P}, \widetilde{\mathbb{P}}$ are *equivalent* if $d\widetilde{\mathbb{P}}/d\mathbb{P}$ is a r.v. ρ with values in $(0, \infty)$.

2.4 Arbitrage

An arbitrage is an opportunity to make risk-free profits in the market. Basic modelling assumption is that no such opportunity exists.

$$\begin{aligned} &(\bar{S}_n)_{0\le n\le T}, \qquad \bar{S}_n = (S^0_n, S_n), \quad S_n = (S^1_n, \dots, S^d_n) \quad \text{asset price model,} \\ &(\bar{\Theta}_n)_{1\le n\le T}, \qquad \bar{\Theta}_n = (\Theta^0_n, \Theta_n), \quad \Theta_n = (\Theta^1_n, \dots, \Theta^d_n) \quad \text{portfolio.} \end{aligned}$$

Value

$$V_0 = \bar{\Theta}_1 \cdot \bar{X}_0, \qquad V_n = \bar{\Theta}_n \cdot \bar{X}_n, \quad n \ge 1, \qquad X^i_n = S^i_n / S^0_n.$$

$$V_n = V_0 + \sum_{k=1}^{n} \Theta_k \cdot (X_k - X_{k-1}).$$

In this set-up an *arbitrage* for $(\bar{S}_n)_{0\le n\le T}$ is a previsible self-financing portfolio with initial value 0 such that

$$V_T \ge 0 \quad \text{a.s.}, \qquad V_T > 0 \quad \text{with positive probability.}$$

Since V_T depends on $(\bar{S}_n)_{0\le n\le T}$ and $(\bar{\Theta}_n)_{1\le n\le T}$ only through

$$V_0 \text{ and } (X_n)_{0\le n\le T} \text{ and } (\Theta_n)_{1\le n\le T},$$

we will often say $(\Theta_n)_{1\le n\le T}$ is an arbitrage for $(X_n)_{0\le n\le T}$.

Consider the case $(X_n)_{0\le n\le T}$ is a martingale in $\mathbb{R}^d$ (all components are mgs) and let $(\Theta_n)_{1\le n\le T}$ be a bounded previsible process in $\mathbb{R}^d$, considered as a portfolio. Then

$$V_n = \sum_{k=1}^{n} \Theta_k \cdot (X_k - X_{k-1}) \qquad \text{is the martingale transform}$$

$$\left(= \sum_{i=1}^{d} \sum_{k=1}^{n} \Theta^i_k (X^i_k - X^i_{k-1}) \right).$$

Suppose Θ chosen so that $V_T \ge 0$ a.s. We have $\mathbb{E}(V_T) = 0$ so $V_T = 0$ a.s.

So we see that $(X_n)_{0\le n\le T}$ has no bounded arbitrage.

Proposition. Let $(X_n)_{0\le n\le T}$ be a martingale in $\mathbb{R}^d$. Then $(X_n)_{0\le n\le T}$ has no arbitrage.

Proof. Let $(\Theta_n)_{1\le n\le T}$ be any previsible process. Let

$$V_n = \sum_{k=1}^{n} \Theta_k \cdot (X_k - X_{k-1}).$$

Suppose $V_T \ge 0$ (WTS $V_T = 0$ a.s.). Note that

$$\mathbb{E}(V_T \mid \mathcal{F}_T) = V_T \quad \text{a.s.}$$

Suppose in a backwards induction for $n \le T$ that

$$\mathbb{E}(V_T \mid \mathcal{F}_n) = V_n \quad \text{a.s.}$$

Note that then $V_n \ge 0$ a.s. Now

$$V_n = V_{n-1} + \Theta_n \cdot (X_n - X_{n-1}).$$

Fix $R < \infty$, set

$$A = \{|V_{n-1}| \le R \text{ and } |\Theta_n| \le R\}.$$

Then A is $\mathcal{F}_{n-1}$-meas. and

$$\mathbf{1}_A V_n = \mathbf{1}_A V_{n-1} + (\Theta_n \mathbf{1}_A) \cdot (X_n - X_{n-1}).$$

Now take $\mathbb{E}(\cdot \mid \mathcal{F}_{n-1})$, taking out what is known,

$$\mathbf{1}_A \mathbb{E}(V_n \mid \mathcal{F}_{n-1}) = \mathbf{1}_A V_{n-1} + (\Theta_n \mathbf{1}_A) \underbrace{\mathbb{E}(X_n - X_{n-1} \mid \mathcal{F}_{n-1})}_{0}.$$

But R was arbitrary and $|\Theta_n|$, $|V_{n-1}|$ are finite-valued, so we must have

$$\mathbb{E}(V_n \mid \mathcal{F}_{n-1}) = V_{n-1} \quad \text{a.s.}$$

so, by tower,

$$\mathbb{E}(V_T \mid \mathcal{F}_{n-1}) = V_{n-1} \quad \text{a.s.}$$

Hence $\mathbb{E}(V_T) = V_0 = 0$ so $V_T = 0$ a.s., so $(\Theta_n)_{1 \le n \le T}$ is not an arbitrage. □

2.5 Fundamental Theorem of Asset Pricing

Theorem. For any asset price model $(\bar{S}_n)_{0 \le n \le T}$ the following are equivalent:

(a) $(\bar{S}_n)_{0 \le n \le T}$ has no arbitrage.

(b) There exists an equivalent probability measure $\widetilde{\mathbb{P}}$ under which the discounted price process $(X_n)_{0 \le n \le T}$ is a martingale.

$$\widetilde{\mathbb{P}} \text{ is called an } \left\{ \begin{array}{l} \text{equivalent martingale} \\ \text{risk-neutral} \end{array} \right\} \text{ measure.}$$

End of lecture 5

Recap of the basic framework

Let $(\bar{S}_n)_{0 \le n \le T}$ be an asset price model in $\mathbb{R}^{d+1}$, with numeraire $S^0_n > 0$, and put

$$X_n = \frac{S_n}{S^0_n}, \qquad \bar{X}_n = (1, X_n).$$

Given an initial discounted wealth V_0 and a previsible process $(\Theta_n)_{1 \le n \le T}$ in $\mathbb{R}^d$, there is an associated self-financing portfolio $(\bar{\Theta}_n)_{1 \le n \le T}$ with

$$V_T = \bar{\Theta}_T \cdot \bar{X}_T = V_0 + \sum_{n=1}^{T} \Theta_n \cdot (X_n - X_{n-1}).$$

Recall that Θ is an arbitrage if

$$V_0 = 0, \qquad V_T \ge 0 \text{ a.s.}, \qquad \mathbb{P}(V_T > 0) > 0.$$

An *equivalent martingale measure* is a probability measure $\widetilde{\mathbb{P}} \sim \mathbb{P}$ under which X is a martingale.

Proof of the Fundamental Theorem of Asset Pricing

The theorem says that the following are equivalent:

(a) X has no arbitrage.

(b) There exists an equivalent martingale measure.

Proof of (b) $\Rightarrow$ (a). By the proposition in Section 2.4, a martingale price process has no arbitrage under $\widetilde{\mathbb{P}}$. Equivalence means that $\mathbb{P}$ and $\widetilde{\mathbb{P}}$ have the same null events, so the conditions defining an arbitrage are the same under either measure. Hence there is no arbitrage under $\mathbb{P}$. $\square$

Proof of (a) $\Rightarrow$ (b) in the single-period case. We prove this direction here when $T = 1$ and $\mathcal{F}_0 = \{\varnothing, \Omega\}$. Write

$$\Theta = \Theta_1 \in \mathbb{R}^d, \qquad Y = X_1 - X_0.$$

Since $\mathcal{F}_0$ is trivial, X_0 and Θ_1 are deterministic (up to null sets). Thus (a) says that there is no $\Theta \in \mathbb{R}^d$ such that

$$\Theta \cdot Y \geq 0 \text{ a.s.}, \qquad \mathbb{P}(\Theta \cdot Y > 0) > 0.$$

Condition (b) asks for $\widetilde{\mathbb{P}} \sim \mathbb{P}$ such that Y is integrable and

$$\widetilde{\mathbb{E}}(Y) = 0.$$

Indeed, this is precisely the martingale condition $\widetilde{\mathbb{E}}(X_1 \mid \mathcal{F}_0) = X_0$ in this setting.

Step 1: ensure finite exponential moments. Set

$$\phi(\Theta) = \mathbb{E}\left(e^{\Theta \cdot Y}\right), \qquad \Theta \in \mathbb{R}^d.$$

We may reduce to the case $\phi(\Theta) < \infty$ for every Θ by first switching to the equivalent probability measure $\widehat{\mathbb{P}}$ given by

$$\frac{d\widehat{\mathbb{P}}}{d\mathbb{P}} = \frac{e^{-|Y|^2}}{\mathbb{E}(e^{-|Y|^2})}.$$

The denominator is strictly positive and finite, and the density is strictly positive. Moreover,

$$\Theta \cdot Y - |Y|^2 = -\left|Y - \frac{\Theta}{2}\right|^2 + \frac{|\Theta|^2}{4} \leq \frac{|\Theta|^2}{4},$$

so

$$\widehat{\mathbb{E}}\left(e^{\Theta \cdot Y}\right) \leq \frac{e^{|\Theta|^2/4}}{\mathbb{E}(e^{-|Y|^2})} < \infty.$$

No arbitrage is preserved under this change of measure. A measure equivalent to $\widehat{\mathbb{P}}$ will also be equivalent to the original $\mathbb{P}$. We therefore relabel $\widehat{\mathbb{P}}$ as $\mathbb{P}$ for the remainder of the proof and assume all exponential moments are finite.

Under this assumption Y is integrable and ϕ is differentiable, with

$$\nabla\phi(\Theta) = \mathbb{E}\left(Y e^{\Theta \cdot Y}\right).$$

To justify differentiation under the expectation, on a bounded set of Θ the derivatives are dominated by an integrable multiple of $e^{a\sum_{i=1}^d |Y^i|}$ for a sufficiently large a. This is integrable because

$$e^{a\sum_{i=1}^d |Y^i|} \leq \sum_{s \in \{-1,1\}^d} e^{as \cdot Y},$$

and every term has finite expectation.

Step 2: reduce the existence of a minimum to relevant directions. We will show that ϕ attains a minimum on $\mathbb{R}^d$. Let

$$E_0 = \{\Theta \in \mathbb{R}^d : \Theta \cdot Y = 0 \text{ a.s.}\}, \qquad E_1 = E_0^{\perp}, \qquad \mathbb{R}^d = E_0 \oplus E_1.$$

For $\Theta_0 \in E_0$ and $\Theta_1 \in E_1$,

$$\phi(\Theta_0 + \Theta_1) = \phi(\Theta_1).$$

Thus it suffices to find a minimum on E_1. If $E_1 = \{0\}$, then $\phi \equiv 1$ and the minimum is already attained. Otherwise, since $\phi(0) = 1$, it will suffice to show that

$$\phi(\Theta) > 1 \qquad \text{for all } \Theta \in E_1 \text{ with } |\Theta| \text{ sufficiently large.}$$

Define

$$\psi(t) = \min\{1, \max\{0, t\}\}, \qquad f(\Theta) = \mathbb{E}(\psi(\Theta \cdot Y)).$$

The function ψ is Lipschitz with constant 1. Also, f is continuous by bounded convergence, since $0 \le \psi \le 1$.

Let

$$\mathcal{S} = \{\Theta \in E_1 : |\Theta| = 1\}.$$

For $\Theta \in \mathcal{S}$ we have $\mathbb{P}(\Theta \cdot Y \neq 0) > 0$, since $E_0 \cap E_1 = \{0\}$. No arbitrage, applied to $-\Theta$, implies

$$\mathbb{P}(\Theta \cdot Y > 0) > 0.$$

Otherwise $\Theta \cdot Y \le 0$ a.s., and $-\Theta$ would give a non-negative gain which is strictly positive with positive probability. It follows that $f(\Theta) > 0$ on $\mathcal{S}$.

Since $\mathcal{S}$ is compact and f is continuous, set

$$\varepsilon = \frac{1}{2} \min_{\Theta \in \mathcal{S}} f(\Theta) > 0.$$

For every $\Theta \in \mathcal{S}$, using the Lipschitz property of ψ,

$$\begin{aligned} \mathbb{P}(\Theta \cdot Y \ge \varepsilon) &\ge \mathbb{E}(\psi(\Theta \cdot Y - \varepsilon)) \\ &\ge \mathbb{E}(\psi(\Theta \cdot Y)) - \varepsilon \\ &= f(\Theta) - \varepsilon \ge \varepsilon. \end{aligned}$$

The first inequality holds because $\psi(z - \varepsilon) \le \mathbf{1}_{\{z \ge \varepsilon\}}$. Consequently, for $t \ge 0$,

$$\begin{aligned} \phi(t\Theta) &= \mathbb{E}\big(e^{t\Theta \cdot Y}\big) \\ &\ge e^{t\varepsilon} \mathbb{P}(\Theta \cdot Y \ge \varepsilon) \ge \varepsilon e^{t\varepsilon} \longrightarrow \infty \qquad (t \to \infty). \end{aligned}$$

The lower bound is uniform over $\Theta \in \mathcal{S}$. Writing any non-zero $\Theta \in E_1$ as $|\Theta|(\Theta/|\Theta|)$ therefore shows that $\phi(\Theta) > 1$ outside a sufficiently large ball in E_1. By continuity, ϕ attains a minimum on the compact ball, hence on E_1 and on $\mathbb{R}^d$. Denote a minimizer by Θ^*.

Step 3: construct the equivalent martingale measure. Set

$$Z = \phi(\Theta^*) \in (0, \infty), \qquad \frac{d\widetilde{\mathbb{P}}}{d\mathbb{P}} = \frac{e^{\Theta^* \cdot Y}}{Z}.$$

This is a strictly positive density with expectation 1, so $\widetilde{\mathbb{P}} \sim \mathbb{P}$. Since Θ^* minimizes the differentiable function ϕ,

$$\nabla \phi(\Theta^*) = \mathbb{E}\Big(Y e^{\Theta^* \cdot Y}\Big) = 0.$$

The same exponential-moment bounds used above give

$$\widetilde{\mathbb{E}}(|Y|) = \frac{\mathbb{E}\big(|Y| e^{\Theta^* \cdot Y}\big)}{Z} < \infty, \qquad \widetilde{\mathbb{E}}(Y) = \frac{\nabla \phi(\Theta^*)}{Z} = 0.$$

For example, coordinatewise integrability follows from $|x| \le e^x + e^{-x}$ and the finiteness of the exponential moments at $\Theta^* + e_i$ and $\Theta^* - e_i$. Thus X is a martingale under $\widetilde{\mathbb{P}}$, proving (b) in the single-period case. $\square$

2.6 Pricing attainable claims

Let C be a time-T contingent claim: C is non-negative and $\mathcal{F}_T$-measurable. Its *discounted value* is

$$D = \frac{C}{S_T^0}.$$

We say that C is *attainable* if there is a previsible self-financing portfolio $(\bar{\Theta}_n)_{1\le n\le T}$ such that

$$C = \bar{\Theta}_T \cdot \bar{S}_T.$$

Equivalently, there exist an $\mathcal{F}_0$-measurable initial discounted wealth V_0 and a previsible process $(\Theta_n)_{1\le n\le T}$ in $\mathbb{R}^d$ such that

$$D = V_0 + \sum_{n=1}^{T} \Theta_n \cdot (X_n - X_{n-1}).$$

Then V_0 is called the *fair price* of C in discounted units, and $\bar{\Theta}$ is called a *replicating* or *hedging portfolio*. In currency units the initial price is $S_0^0 V_0$.

Proposition. Assume that

$$\mathcal{F}_0 = \{\varnothing, \Omega\}, \qquad \mathcal{F}_T = \sigma(\bar{S}_1, \dots, \bar{S}_T).$$

Let C be an attainable contingent claim and let $\widetilde{\mathbb{P}}$ be any equivalent martingale measure. Then

$$\boxed{V_0 = \widetilde{\mathbb{E}}\left(\frac{C}{S_T^0}\right).}$$

Proof. Since C is attainable, there is a previsible process Θ with

$$D = \frac{C}{S_T^0} = V_0 + \sum_{n=1}^{T} \Theta_n \cdot (X_n - X_{n-1}).$$

Put

$$V_n = V_0 + \sum_{k=1}^{n} \Theta_k \cdot (X_k - X_{k-1}), \qquad 0 \le n \le T,$$

so that $V_T = D \ge 0$. We spell out the backwards conditional-expectation argument from the proposition in Section 2.4. At time T,

$$\widetilde{\mathbb{E}}[D \mid \mathcal{F}_T] = V_T \ge 0.$$

Suppose that

$$\widetilde{\mathbb{E}}[D \mid \mathcal{F}_{n+1}] = V_{n+1} \ge 0.$$

To justify taking out the possibly unbounded holdings, fix $R < \infty$ and set

$$A_R = \{|V_n| \le R,\ |\Theta_{n+1}| \le R\} \in \mathcal{F}_n.$$

The identity

$$\mathbf{1}_{A_R} V_{n+1} = \mathbf{1}_{A_R} V_n + (\mathbf{1}_{A_R}\Theta_{n+1}) \cdot (X_{n+1} - X_n)$$

shows that $\mathbf{1}_{A_R} V_{n+1}$ is integrable. Since X is a martingale under $\widetilde{\mathbb{P}}$,

$$\begin{aligned} \mathbf{1}_{A_R}\widetilde{\mathbb{E}}[V_{n+1} \mid \mathcal{F}_n] &= \mathbf{1}_{A_R} V_n + (\mathbf{1}_{A_R}\Theta_{n+1}) \cdot \widetilde{\mathbb{E}}[X_{n+1} - X_n \mid \mathcal{F}_n] \\ &= \mathbf{1}_{A_R} V_n. \end{aligned}$$

Letting $R \to \infty$ gives

$$\widetilde{\mathbb{E}}[V_{n+1} \mid \mathcal{F}_n] = V_n \geq 0.$$

Since $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$, the tower property and the backwards-induction hypothesis give

$$\begin{aligned}\widetilde{\mathbb{E}}[D \mid \mathcal{F}_n] &= \widetilde{\mathbb{E}}\Big[\widetilde{\mathbb{E}}[D \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\Big] \\ &= \widetilde{\mathbb{E}}[V_{n+1} \mid \mathcal{F}_n] \\ &= V_n.\end{aligned}$$

Thus, by backwards induction,

$$\widetilde{\mathbb{E}}(D \mid \mathcal{F}_n) = V_n, \qquad 0 \leq n \leq T.$$

At time 0, V_0 is deterministic and finite, so

$$\widetilde{\mathbb{E}}(D) = V_0.$$

In particular, D and all the V_n are integrable under $\widetilde{\mathbb{P}}$. Hence the discounted wealth process is a martingale, with $V_0 = \widetilde{\mathbb{E}}(C/S_T^0)$ as claimed. □

Interpretation. For an attainable claim, every equivalent martingale measure gives the same discounted expected payoff: it equals the initial cost of a replicating portfolio.

End of lecture 6

2.8 Completeness

An asset price model $(\bar{S}_n)_{0 \leq n \leq T}$ is called *complete* if every time-T contingent claim C is attainable. Equivalently, for every such C there exist an initial discounted wealth V_0 and a previsible process $(\Theta_n)_{1 \leq n \leq T}$ such that

$$\frac{C}{S_T^0} = V_0 + \sum_{n=1}^{T} \Theta_n \cdot (X_n - X_{n-1}).$$

Proposition. Assume that

$$\mathcal{F}_0 = \{\varnothing, \Omega\}, \qquad \mathcal{F}_T = \sigma(\bar{S}_0, \ldots, \bar{S}_T),$$

and that the numeraire $(S_n^0)_{0 \leq n \leq T}$ is non-random. If the model is complete, then there is at most one equivalent martingale measure.

Proof. Let $A \in \mathcal{F}_T$. The indicator $\mathbf{1}_A$ is a time-T contingent claim, so completeness makes it attainable. Write its fair value in discounted units as

$$V_0(A) = \frac{\nu(A)}{S_T^0}.$$

For any equivalent martingale measure $\widetilde{\mathbb{P}}$, the pricing proposition from Section 2.6 gives

$$\frac{\widetilde{\mathbb{P}}(A)}{S_T^0} = \widetilde{\mathbb{E}}\left(\frac{\mathbf{1}_A}{S_T^0}\right) = V_0(A) = \frac{\nu(A)}{S_T^0}.$$

Hence $\widetilde{\mathbb{P}}(A) = \nu(A)$ for every $A \in \mathcal{F}_T$, so the equivalent martingale measure, if it exists, is unique on $\mathcal{F}_T$. □

2.9 Binomial model

Fix parameters

$$r, a, b \in (-1, \infty), \qquad a < b, \qquad p \in (0,1), \qquad S_0 \in (0, \infty).$$

Here $d = 1$. The model $(S_n^0, S_n)_{0 \le n \le T}$ is a *binomial model* with interest rate r and parameters a, b, p if

$$S_n^0 = (1+r)^n, \qquad S_n = S_0 \prod_{i=1}^{n} (1 + R_i),$$

where $R_1, \dots, R_T$ are independent and identically distributed, with

$$\mathbb{P}(R_n = a) = 1 - p, \qquad \mathbb{P}(R_n = b) = p.$$

We use the natural filtration $\mathcal{F}_n = \sigma(S_0, \dots, S_n)$ and write $X_n = S_n / S_n^0$ for the discounted risky asset price.

Proposition. A binomial model has arbitrage unless $r \in (a, b)$.
Proof. Consider the self-financing portfolio with discounted initial wealth $V_0 = 0$ and risky holdings

$$\Theta_1 = 1, \qquad \Theta_n = 0 \quad (n \ge 2).$$

Its terminal discounted wealth is

$$V_T = X_1 - X_0 = \frac{S_0(1+R_1)}{1+r} - S_0 = \frac{S_0(R_1 - r)}{1+r}.$$

If $r \le a$, then $V_T \ge 0$ and $V_T > 0$ on $\{R_1 = b\}$, which has probability $p > 0$. Thus this portfolio is an arbitrage. If $r \ge b$, then $-V_T \ge 0$ and $-V_T > 0$ on $\{R_1 = a\}$, which has probability $1 - p > 0$. The opposite portfolio is therefore an arbitrage. □

The risk-neutral measure

Proposition. Suppose that $r \in (a, b)$ and define

$$p^* = \frac{r - a}{b - a} \in (0,1), \qquad r = (1 - p^*)a + p^* b.$$

Let

$$U_T = \#\{1 \le n \le T : R_n = b\}, \qquad D_T = \#\{1 \le n \le T : R_n = a\} = T - U_T,$$

and define $\mathbb{P}^*$ by

$$\frac{d\mathbb{P}^*}{d\mathbb{P}} = \left(\frac{p^*}{p}\right)^{U_T} \left(\frac{1-p^*}{1-p}\right)^{D_T}.$$

Then $\mathbb{P}^*$ is an equivalent martingale measure. Under $\mathbb{P}^*$ the model is binomial with up probability p^*. In particular, the binomial model has no arbitrage when $r \in (a, b)$.
Proof. For a fixed return path $(r_1, \dots, r_T) \in \{a, b\}^T$, put

$$u = \#\{1 \le n \le T : r_n = b\}, \qquad d = T - u.$$

Then

$$\begin{aligned} \mathbb{P}^*(R_1 = r_1, \dots, R_T = r_T) &= (1-p)^d p^u \left(\frac{p^*}{p}\right)^u \left(\frac{1-p^*}{1-p}\right)^d \\ &= (1-p^*)^d (p^*)^u. \end{aligned}$$

These probabilities sum to 1, and the density is strictly positive. Thus $\mathbb{P}^* \sim \mathbb{P}$, and under $\mathbb{P}^*$ the returns are i.i.d. with up probability p^*. Moreover,

$$\mathbb{E}^*\left(\frac{1+R_n}{1+r}\right) = \frac{(1+a)(1-p^*)+(1+b)p^*}{1+r} = \frac{1+(1-p^*)a+p^*b}{1+r} = 1.$$

Since

$$X_n = S_0 \prod_{i=1}^{n} \frac{1+R_i}{1+r},$$

independence gives

$$\mathbb{E}^*(X_n \mid \mathcal{F}_{n-1}) = X_{n-1}.$$

The process X is integrable (there are only finitely many return paths), so it is a martingale under $\mathbb{P}^*$, as required. The Fundamental Theorem of Asset Pricing implies no arbitrage. □

Pricing claims in the binomial model

The binomial model will also be shown to be complete. For now, suppose that a time-T claim $C = f(S_T)$ is attainable, with discounted fair value V_0. Then

$$\begin{aligned} V_0 &= \mathbb{E}^*\left(\frac{C}{S_T^0}\right) \\ &= \frac{1}{(1+r)^T} \sum_{k=0}^{T} \binom{T}{k} (1-p^*)^{T-k} (p^*)^k f\Big(S_0(1+a)^{T-k}(1+b)^k\Big). \end{aligned}$$

More generally, for an attainable path-dependent claim $C = f(S_0, S_1, \ldots, S_T)$,

$$V_0 = \frac{1}{(1+r)^T} \sum_{\text{all price paths } (s_0,\ldots,s_T)} f(s_0,\ldots,s_T)(1-p^*)^d (p^*)^u,$$

where $s_0 = S_0$ and

$$u = \#\{1 \le n \le T : s_n = (1+b)s_{n-1}\}, \qquad d = T - u.$$

The sum runs over the 2^T possible price paths starting from S_0.

Backward recursion on the price tree

The path sum can be organized as a backward recursion on a binary tree. Set

$$f_T(s_0,\ldots,s_T) = f(s_0,\ldots,s_T).$$

For $n = T-1, \ldots, 0$, define

$$\begin{aligned} f_n(s_0,\ldots,s_n) = (1-p^*) f_{n+1}(s_0,\ldots,s_n,(1+a)s_n) \\ + p^* f_{n+1}(s_0,\ldots,s_n,(1+b)s_n). \end{aligned}$$

At each node, this averages the two successor values with their risk-neutral probabilities. The functions f_n give expected terminal payoffs; the discount factor is applied separately.

Proposition. For $n = 0, 1, \ldots, T$,

$$\mathbb{E}^*[f(S_0,\ldots,S_T) \mid \mathcal{F}_n] = f_n(S_0,\ldots,S_n) \qquad \text{a.s.}$$

In particular,

$$\mathbb{E}^*(C) = f_0(S_0), \qquad V_0 = \frac{f_0(S_0)}{(1+r)^T}.$$

Proof. The assertion holds for $n = T$. Suppose inductively that it holds for $n+1$, where $n \le T-1$. Fix a possible price history $(s_0, \dots, s_n)$ and let

$$A = \{S_0 = s_0, \dots, S_n = s_n\}.$$

By the tower property and the induction hypothesis,

$$\begin{aligned}
\mathbb{E}^*[f(S_0, \dots, S_T) \mid A] &= \mathbb{E}^*[f_{n+1}(S_0, \dots, S_{n+1}) \mid A] \\
&= (1-p^*) f_{n+1}(s_0, \dots, s_n, (1+a)s_n) \\
&\quad + p^* f_{n+1}(s_0, \dots, s_n, (1+b)s_n) \\
&= f_n(s_0, \dots, s_n) \\
&= \mathbb{E}^*[f_n(S_0, \dots, S_n) \mid A]\,.
\end{aligned}$$

Now $f_n(S_0, \dots, S_n)$ is $\mathcal{F}_n$-measurable, and every event in $\mathcal{F}_n$ is a finite union of such history events. Thus the defining property of conditional expectation gives the claim at time n, completing the backward induction. □

End of lecture 7

2.10 Completeness of the binomial model

Recall that

$$S_n^0 = (1+r)^n, \qquad S_n = S_0 \prod_{i=1}^{n} (1+R_i),$$

where $R_i = b$ with probability p and $R_i = a$ with probability $1-p$. Assume $a < r < b$ and set $p^* = (r-a)/(b-a)$, so that $\mathbb{P}^*$ is an equivalent martingale measure. For a contingent claim $C = f(S_0, \dots, S_T)$, the backward recursion from Section 2.9 gives

$$\mathbb{E}^*(C \mid \mathcal{F}_n) = f_n(S_0, \dots, S_n) \qquad \text{a.s.},$$

where $\mathcal{F}_n = \sigma(R_1, \dots, R_n)$, $f_T = f$, and

$$\begin{aligned}
f_n(s_0, \dots, s_n) = (1-p^*) f_{n+1}(s_0, \dots, s_n, (1+a)s_n) \\
+ p^* f_{n+1}(s_0, \dots, s_n, (1+b)s_n), \qquad n < T.
\end{aligned}$$

For $1 \le n \le T$, define

$$\Delta_n(s_0, \dots, s_{n-1}) = \frac{f_n(s_0, \dots, s_{n-1}, (1+b)s_{n-1}) - f_n(s_0, \dots, s_{n-1}, (1+a)s_{n-1})}{(1+r)^{T-n}(b-a)s_{n-1}}.$$

Proposition 2.10. The contingent claim $C = f(S_0, \dots, S_T)$ has a replicating portfolio whose risky holdings are

$$\Theta_n = \Delta_n(S_0, \dots, S_{n-1}), \qquad 1 \le n \le T.$$

In particular, the binomial model is complete.

Proof. Set

$$V_n = \mathbb{E}^*\left(\frac{C}{S_T^0} \,\middle|\, \mathcal{F}_n \right) = \frac{f_n(S_0, \dots, S_n)}{(1+r)^T}.$$

Fix a possible price history $(s_0, \dots, s_{n-1})$, starting from $s_0 = S_0$, and define

$$\begin{aligned}
\Omega_0 &= \{S_1 = s_1, \dots, S_{n-1} = s_{n-1}\}, \\
\Omega_a &= \Omega_0 \cap \{R_n = a\}, \qquad \Omega_b = \Omega_0 \cap \{R_n = b\}.
\end{aligned}$$

For $n = 1$, take $\Omega_0 = \Omega$. Write

$$\phi(x) = f_n(s_0, \dots, s_{n-1}, (1+x)s_{n-1}).$$

On Ω_a,

$$V_n = \frac{\phi(a)}{(1+r)^T}, \qquad V_{n-1} = \frac{(1-p^*)\phi(a) + p^*\phi(b)}{(1+r)^T},$$

and hence

$$V_n - V_{n-1} = \frac{p^*(\phi(a) - \phi(b))}{(1+r)^T}.$$

For the discounted stock price $X_n = S_n/(1+r)^n$, we have on the same event

$$X_n - X_{n-1} = \left(\frac{1+a}{1+r} - 1\right) X_{n-1} = \frac{(a-r)s_{n-1}}{(1+r)^n}.$$

Consequently,

$$\begin{aligned}\Theta_n(X_n - X_{n-1}) &= \frac{\phi(b) - \phi(a)}{(1+r)^{T-n}(b-a)s_{n-1}} \frac{(a-r)s_{n-1}}{(1+r)^n} \\ &= \frac{p^*(\phi(a) - \phi(b))}{(1+r)^T} = V_n - V_{n-1},\end{aligned}$$

using $(a-r)/(b-a) = -p^*$. A similar calculation on Ω_b uses $(b-r)/(b-a) = 1-p^*$ and gives

$$\Theta_n(X_n - X_{n-1}) = \frac{(1-p^*)(\phi(b) - \phi(a))}{(1+r)^T} = V_n - V_{n-1}.$$

Since the history was arbitrary, this identity holds everywhere. The holdings Θ_n are $\mathcal{F}_{n-1}$-measurable, and summing the increments yields

$$\frac{C}{S_T^0} = V_T = V_0 + \sum_{n=1}^{T} \Theta_n(X_n - X_{n-1}).$$

Thus the self-financing portfolio with initial discounted wealth V_0 and risky holdings $(\Theta_n)_{1 \le n \le T}$ replicates C. □

2.11 American calls and puts

An *American call* with expiry T confers the right, but not the obligation, to buy one unit of the underlying asset $(S_n)_{0\leq n\leq T}$ for the strike price K at any stopping time $\tau \leq T$. Its payoff on exercise is

$$C_\tau = (S_\tau - K)^+.$$

An *American put* instead confers the right to sell one unit for K, with exercise payoff

$$C_\tau = (K - S_\tau)^+.$$

These are not simply time-T claims: the holder chooses when to exercise. We seek to optimize the discounted exercise payoff C_τ / S^0_τ.

Assume that the asset price model is complete and arbitrage-free. Then, for any two stopping times $\tau_1, \tau_2 \leq T$, the difference of discounted payoffs

$$\frac{C_{\tau_1}}{S^0_{\tau_1}} - \frac{C_{\tau_2}}{S^0_{\tau_2}}$$

can be replicated in the market, with initial price in discounted units

$$\mathbb{E}^*\left(\frac{C_{\tau_1}}{S^0_{\tau_1}}\right) - \mathbb{E}^*\left(\frac{C_{\tau_2}}{S^0_{\tau_2}}\right).$$

It therefore suffices to maximize

$$\mathbb{E}^*\left(\frac{C_\tau}{S^0_\tau}\right)$$

over stopping times $\tau \leq T$.

End of lecture 8

3. Brownian motion

3.1 Definition and basic properties

A real-valued process $(B_t)_{t\geq 0}$, with $t \in [0,\infty)$ and $B_0 = 0$, is called a *Brownian motion* if:

(a) For $s, t \geq 0$, $B_{s+t} - B_s$ has $N(0,t)$ distribution and is independent of

$$\sigma(B_r : r \leq s) = \mathcal{F}_s.$$

(b) For all $\omega \in \Omega$, the map

$$t \longmapsto B_t(\omega) : [0,\infty) \longrightarrow \mathbb{R}$$

is continuous. Say B is continuous.

Sometimes we replace $B_0 = 0$ by $B_0 = x$. Then B is a BM starting from x. We'll write $\mathbb{P}_x, \mathbb{E}_x$ as a reminder.

Proposition 3.1. Let $B = (B_t)_{t\geq 0}$ be a continuous real-valued process. Then the following are equivalent:

(a) B is a Brownian motion.

(b) B is a zero-mean Gaussian process with covariance

$$\mathbb{E}(B_s B_t) = s \wedge t.$$

Proof. Suppose (a) holds. Then $\mathbb{E}(B_t) = 0$ for all t. Also, for $0 = t_0 < t_1 < \cdots < t_n$,

$$B_{t_1} - B_{t_0}, \ldots, B_{t_n} - B_{t_{n-1}}$$

are independent Gaussian random variables, with $B_{t_0} = 0$. So $(B_{t_1}, \ldots, B_{t_n})$ is also Gaussian. For $s \leq t$, we have

$$\begin{aligned}\mathbb{E}(B_s B_t) &= \mathbb{E}(B_s^2) + \underbrace{\mathbb{E}\big(B_s(B_t - B_s)\big)}_{=0} \\ &= s = s \wedge t.\end{aligned}$$

So (b) holds.

Assume (b) holds. Then, for $s, t \geq 0$ and $r \leq s$,

$$\mathbb{E}(B_{s+t} - B_s) = 0,$$

and

$$\mathbb{E}\big((B_{s+t} - B_s)^2\big) = (t+s) - 2s + s = t.$$

So $B_{s+t} - B_s \sim N(0,t)$. Also,

$$\mathbb{E}\big((B_{s+t} - B_s)B_r\big) = (t+s) \wedge r - s \wedge r = 0.$$

Since B is Gaussian, this shows that $B_{s+t} - B_s$ is independent of

$$(B_{r_1}, \ldots, B_{r_n}) \qquad \text{for all } r_1, \ldots, r_n \leq s. \qquad \square$$

Exercise (Scaling property). Let B be a BM. Let $c > 0$. Set

$$B_t^{(c)} = c^{-1} B_{c^2 t}.$$

Then $(B_t^{(c)})_{t\geq 0}$ is also a BM.

Proposition. Let B be a BM. Then B leaves every finite interval a.s.
Proof. Fix $L > 0$ and consider the events

$$A_n = \{|B_n - B_{n-1}| > L\}.$$

Then $A_1, A_2, \ldots$ are independent, with

$$\mathbb{P}(A_n) = p = \mathbb{P}(|Z| > L) > 0, \qquad Z \sim N(0,1).$$

So

$$\mathbb{P}\left(\bigcap_n A_n^c\right) = \prod_n (1-p) = 0.$$

But on $\bigcup_n A_n$, B eventually jumps by L between $n-1$ and n. So B leaves any interval of length $< L$ eventually. □

$$\mathcal{F}_t = \sigma(B_s : s \le t).$$

Say $T : \Omega \longrightarrow [0, \infty]$ is a *stopping time* if

$$\{T \le t\} \in \mathcal{F}_t \qquad \text{for all } t \ge 0.$$

Set

$$\mathcal{F}_T = \{A \in \mathcal{F}_\infty : A \cap \{T \le t\} \in \mathcal{F}_t \text{ for all } t \ge 0\},$$

where $\mathcal{F}_\infty = \sigma(B_t : t \ge 0)$. This is what we know by time T.

Proposition (Strong Markov property). Let B be a BM and let T be a finite-valued stopping time. Set

$$\widehat{B}_t = B_{T+t} - B_T.$$

Then $(\widehat{B}_t)_{t \ge 0}$ is also a BM and it is independent of $\mathcal{F}_T$ (in particular, of B_T).

Proposition 3.5. Let $(B_t)_{t \ge 0}$ be a Brownian motion and let $a \in \mathbb{R}$. Define

$$T_a = \inf\{t \ge 0 : B_t = a\}.$$

Then T_a is an almost surely finite stopping time.
Proof. It suffices to consider the case $a > 0$. Since $(B_t)_{t \ge 0}$ is continuous, we have

$$\{T_a \le t\} = \{B_t = a\} \cup \bigcap_{\substack{r < a \\ r \in \mathbb{Q}}} \bigcup_{\substack{0 \le s < t \\ s \in \mathbb{Q}}} \{B_s > r\} \in \mathcal{F}_t.$$

Hence T_a is a stopping time.

Set $T = T_1 \wedge T_{-1}$. Then T is a stopping time and, since Brownian motion leaves every interval, $T < \infty$ almost surely. By symmetry

$$\mathbb{P}(B_T = 1) = \mathbb{P}(B_T = -1) = \frac{1}{2}.$$

Then, by the strong Markov property, the sequence of integers hit by $(B_t)_{t \ge 0}$, omitting immediate repeats, is a simple symmetric random walk. Since the random walk is recurrent, it follows that $T_a < \infty$ almost surely. □

3.2 Wiener's theorem

$$W = C[0,\infty) = C([0,\infty),\mathbb{R}), \qquad \mathcal{W} = \sigma(X_t : t \geq 0).$$
$$X_t : W \longrightarrow \mathbb{R}, \qquad X_t(w) = w(t) \quad \text{(coordinate functions)}.$$

Theorem. There is a unique probability measure μ on $(W, \mathcal{W})$ such that $(X_t)_{t\geq 0}$ is a BM_0.
Proved in Advanced Probability.

If B is any BM, we consider the W-valued random variable

$$B(\omega)(t) = B_t(\omega) \in W.$$

Then the law of B on W makes $(X_t)_{t\geq 0}$ into a BM. So the construction of any BM shows existence in Wiener's theorem. Uniqueness in Wiener's theorem shows all BMs have the same law on W.

End of lecture 9

Recap. A real-valued process $(B_t)_{t\geq 0}$ starting at 0 is a Brownian motion if, for all $s, t \geq 0$,

$$B_{t+s} - B_s \sim N(0,t) \quad \text{and is independent of } \mathcal{F}_s^B,$$

and every sample path $t \mapsto B_t(\omega)$ is continuous.

3.3 Brownian motion as a limit of random walks

A discrete-time process $(W_n)_{n\geq 0}$ is a *random walk* if

$$W_0 = 0, \qquad W_n = Y_1 + \cdots + Y_n,$$

where $Y_1, Y_2, \ldots$ are independent and identically distributed.

Theorem (Donsker's invariance principle). Let $(W_n)_{n\geq 0}$ be a random walk in $\mathbb{R}$ whose steps have mean 0 and variance 1. Extend it to a continuous process by linear interpolation:

$$W_{n+u} = (1-u)W_n + uW_{n+1}, \qquad n \geq 0, \quad u \in [0,1].$$

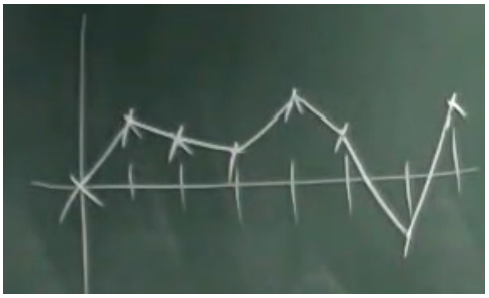
For each integer $k \geq 1$, set

$$W_t^{(k)} = k^{-1/2} W_{kt}, \qquad t \geq 0.$$

Then, for every bounded continuous functional $F : C[0,\infty) \to \mathbb{R}$,

$$\mathbb{E}\big(F(W^{(k)})\big) \longrightarrow \mathbb{E}\big(F(B)\big) \qquad \text{as } k \to \infty,$$

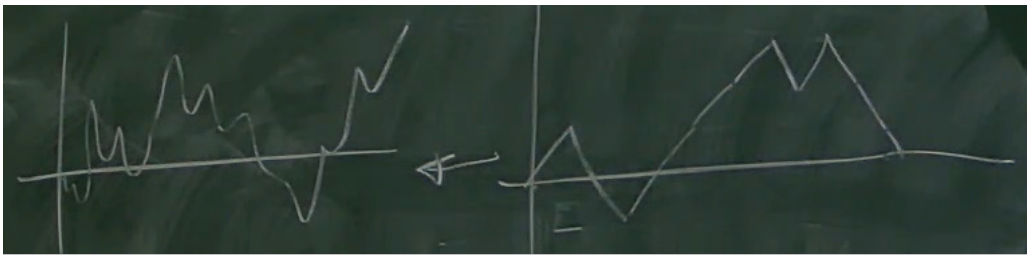
where B is a BM_0.



Here $C[0,\infty)$ is equipped with the topology of uniform convergence on compact intervals, induced by the metric

$$d(w_1, w_2) = \sum_{n=1}^{\infty} 2^{-n} \left(\sup_{t\in[0,n]} |w_1(t) - w_2(t)| \wedge 1 \right).$$

Thus $W^{(k)}$ converges in distribution as a random continuous path. The finite-variance assumption on the steps is needed.



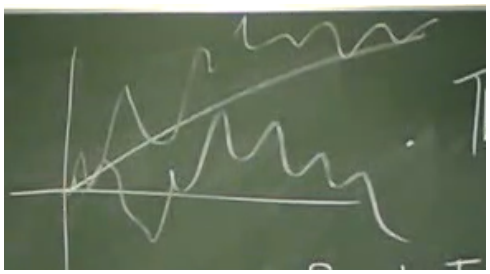
3.4 Change of measure for Brownian motion

Proposition. Let $(B_t)_{0\le t\le T}$ be a BM_0, where $T \ge 0$, and let $c \in \mathbb{R}$. Define

$$\widetilde{B}_t = B_t + ct, \qquad 0 \le t \le T.$$

Then, for every Borel set $A \subseteq C[0,T]$,

$$\boxed{\mathbb{P}(\widetilde{B} \in A) = \mathbb{E}\Big(\mathbf{1}_{\{B\in A\}} e^{cB_T - \frac{1}{2}c^2T}\Big).}$$



Proof. The case $T = 0$ is immediate, so suppose $T > 0$. By a standard π-system argument, it suffices to prove the identity for sets of the form

$$A = \{x \in C[0,T] : x(t_k) - x(t_{k-1}) \in I_k \text{ for } k = 1,\dots,n\},$$

where $0 = t_0 < t_1 < \cdots < t_n = T$ and $I_1,\dots,I_n$ are intervals. Both laws are supported on paths starting at 0; on that subspace, these sets generate the Borel σ-algebra.

First suppose $X \sim N(0,s)$, with $s > 0$, and set $\widehat{X} = X + cs \sim N(cs,s)$. For any interval I,

$$\begin{aligned}
\mathbb{P}(\widehat{X} \in I) &= \int_I \frac{1}{\sqrt{2\pi s}} \exp\left(-\frac{(x-cs)^2}{2s}\right) dx \\
&= \int_I \frac{1}{\sqrt{2\pi s}} e^{-x^2/(2s)} e^{cx - \frac{1}{2}c^2 s}\, dx \\
&= \mathbb{E}\Big(\mathbf{1}_{\{X\in I\}} e^{cX - \frac{1}{2}c^2 s}\Big).
\end{aligned}$$

Set

$$s_k = t_k - t_{k-1}, \qquad X_k = B_{t_k} - B_{t_{k-1}}.$$

Then $X_1,\dots,X_n$ are independent, with $X_k \sim N(0,s_k)$, and

$$\widehat{X}_k := \widetilde{B}_{t_k} - \widetilde{B}_{t_{k-1}} = X_k + cs_k.$$

Consequently,

$$\begin{aligned}
\mathbb{P}(\widetilde{B} \in A) &= \prod_{k=1}^n \mathbb{P}(\widehat{X}_k \in I_k) \\
&= \prod_{k=1}^n \mathbb{E}\Big(\mathbf{1}_{\{X_k\in I_k\}} e^{cX_k - \frac{1}{2}c^2 s_k}\Big) \\
&= \mathbb{E}\Big(\mathbf{1}_{\{B\in A\}} e^{cB_T - \frac{1}{2}c^2 T}\Big),
\end{aligned}$$

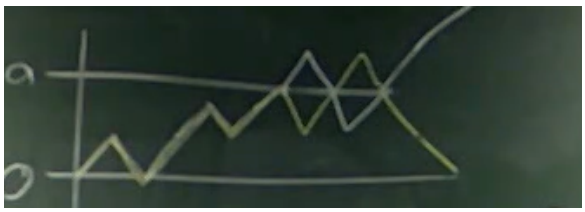
using $\sum_{k=1}^n X_k = B_T$ and $\sum_{k=1}^n s_k = T$. This proves the result. □

3.5 The reflection principle

Theorem. Let $(B_t)_{t\geq 0}$ be a BM_0, and let $a \geq 0$. Set

$$T = \inf\{t \geq 0 : B_t = a\}, \qquad \widetilde{B}_t = \begin{cases} B_t, & t \leq T, \\ a - (B_t - a) = 2a - B_t, & t > T. \end{cases}$$

Then $(\widetilde{B}_t)_{t\geq 0}$ is also a BM_0.



Proof. We know that T is a stopping time and is almost surely finite. Set

$$Y_t = B_{t\wedge T}, \qquad Z_t = B_{T+t} - a, \qquad t \geq 0.$$

By the strong Markov property, $Z = (Z_t)_{t\geq 0}$ is a BM_0 independent of $\mathcal{F}_T^B$. The stopped path Y is $\mathcal{F}_T^B$-measurable, so Y and Z are independent. Since Z and $-Z$ have the same law,

$$(Y, Z) \stackrel{d}{=} (Y, -Z).$$

To reconstruct the full path, let

$$\tau(y) = \inf\{t \geq 0 : y(t) = a\}$$

and define the concatenation map

$$F(y, z)(t) = \begin{cases} y(t), & t \leq \tau(y), \\ a + z(t - \tau(y)), & t > \tau(y). \end{cases}$$

Here $y \in C[0, \infty)$ and $z \in C_0[0, \infty) := \{z \in C[0, \infty) : z(0) = 0\}$. If $\tau(y) = \infty$, use the first branch for every t. The map

$$F : C[0, \infty) \times C_0[0, \infty) \longrightarrow C[0, \infty)$$

is measurable. Since

$$B = F(Y, Z), \qquad \widetilde{B} = F(Y, -Z),$$

B and $\widetilde{B}$ have the same law. □

3.6 The maximum of Brownian motion

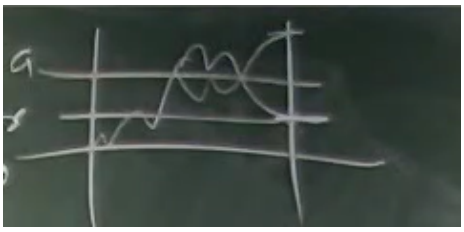
Set

$$M_t = \sup_{0 \le s \le t} B_s.$$

For $a \ge 0$ and $x \le a$, reflection at the first hitting time of a gives the pathwise identity

$$\{M_t \ge a,\ B_t \le x\} = \{\widetilde{B}_t \ge 2a - x\}.$$

Indeed, on $\{M_t \ge a\}$ the path has hit a by time t, and its reflected endpoint is $2a - B_t$. Conversely, an endpoint at least $2a - x \ge a$ forces a hit of a.



Thus

$$\mathbb{P}(M_t \ge a,\ B_t \le x) = \mathbb{P}(\widetilde{B}_t \ge 2a - x) = \mathbb{P}(B_t \ge 2a - x).$$

In particular, for $t > 0$,

$$\begin{aligned} \mathbb{P}(M_t \ge a) &= \mathbb{P}(M_t \ge a,\ B_t \le a) + \mathbb{P}(B_t > a) \\ &= 2\mathbb{P}(B_t \ge a). \end{aligned}$$

Since B_t has a symmetric distribution, it follows that

$$\boxed{M_t \overset{d}{=} |B_t|.}$$

End of lecture 10